Stephen Purdue

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EDUCATION

Royal Holloway, University of London — **London, UK**
Master of Science - Artificial Intelligence — *2026 - Present*

University of Winchester — **Winchester, UK**
Bachelor of Science - Computer Game Development — *2023 - 2026*

WORK EXPERIENCE

Sous-Chef, Chef de Partie
Mustang Group, Southampton, UK — *2021 - Present*

- Led a kitchen of five through high-volume, deadline-driven service under constant time pressure, requiring rapid problem-solving, attention to detail, and direct communication with other colleagues in an agile environment.
- Mentored and trained 4-8 junior chefs and new starters per intake, maintaining consistent service standards and reducing onboarding time during high staff turnover.
- Maintained stock rotation and portion-control procedures to reduce food waste and control ingredient costs across a high-volume kitchen.

PROJECTS

- **Stock Market Prediction Engine**, Developed an LSTM model in PyTorch to forecast daily stock prices from historical time-series analysis data, reducing prediction error by 81% versus a moving-average baseline. Containerised the full stack with Docker Compose, reducing local setup time to under 5 minutes. GitHub
- **Adaptive Combat Simulation**, Developed an adaptive combat simulator, achieving a 75% win rate against a baseline agent, with overall reward improving from 32% to 87%, measured over 30 unique evaluation simulations by training a PPO-agent using Python and TensorFlow. GitHub
- **Procedural Generation Algorithm**, Engineered a scalable procedural generation system in Unity using C# and Binary Space Partitioning, to recursively subdivide space into room layouts and connective corridors in linear time. Reduced level build-time by 77% by implementing modular, parameter-driven logic. GitHub
- **Ad Trigger Discovery Engine**, Built the analysis backend for an ad-tech pipeline, generating synthetic user prompts across intent buckets and running BM25/embedding-based retrieval to surface and rank high-intent keyword triggers from collected brand data, with full source traceability for review. GitHub

TECHNICAL SKILLS

Programming Languages: Python, C#, C++, SQL, Bash
Frameworks and Tools: Linux, Git, Docker, CI/CD, PyTorch, TensorFlow
Cloud: AWS (S3, SageMaker, Bedrock)

INTERESTS

- Snowboarding, Competitive Gaming